

AI / Data Science

Task Report — Day 03

Auto MPG Dataset · Regression · Linear Regression vs Random Forest

DATE

14 June 2026

DOMAIN

Artificial Intelligence / Data Science

DATASET

Auto MPG Dataset (398 rows x 9 cols)

1 Task Overview

#	Objective	Status
1	Select a dataset and perform complete preprocessing	✓ Done
2	Train at least two machine learning models	✓ Done
3	Compare model performance using evaluation metrics	✓ Done
4	Create a dashboard/report with visualizations	✓ Done
5	Document observations and conclusions	✓ Done

2 Dataset — Auto MPG

Source: seaborn-data (GitHub) **Rows:** 398 **Columns:** 9 **Task:** Regression — predict fuel efficiency (mpg)

Column	Description	Notes
mpg	Miles per gallon — target variable	Target
cylinders	Number of engine cylinders	Numeric
displacement	Engine size	Numeric
horsepower	Engine power	Had 6 missing
weight	Car weight (lbs)	Numeric
acceleration	0-60 mph time	Numeric
model_year	Year of manufacture (70-82)	Numeric
origin	Country: usa / europe / japan	Categorical
name	Car model name	Dropped

3 Preprocessing

The dataset required minimal cleaning. The horsepower column had 6 missing values (originally marked as '?'), and the name column had too many unique values to be useful for modelling. The categorical origin column was encoded into numbers so it could be used by the machine learning models.

Step	Action	Reason
1	Filled missing horsepower with median	6 rows had no horsepower value
2	Dropped name column	Too many unique values — not useful
3	Encoded origin (usa=0, europe=1, japan=2)	Models require numeric input

```
df['horsepower'] = df['horsepower'].fillna(df['horsepower'].median())
df = df.drop(columns=['name'])
df['origin_num'] = df['origin'].map({'usa':0, 'europe':1, 'japan':2})
```

4 Feature Engineering

A new feature, power_to_weight, was created by dividing horsepower by weight. This single ratio captures how much engine power a car has relative to its size — often a better efficiency indicator than either value alone.

```
df['power_to_weight'] = df['horsepower'] / df['weight']
```

New Column	Formula	Purpose
power_to_weight	horsepower / weight	Captures engine efficiency relative to car size

5 Visualizations (Dashboard)

Chart 1

MPG Distribution (Histogram) → chart1_mpg_distribution.png

Insight

Most cars in the dataset get between 15-30 mpg, with an average around 23.5 mpg — a fairly normal spread.

Chart 2

Weight vs MPG (Scatter Plot) → chart2_weight_vs_mpg.png

Insight

A strong negative relationship — as car weight increases, fuel efficiency drops sharply. This is the clearest pattern in the dataset.

Chart 3

MPG by Origin (Box Plot) → chart3_mpg_by_origin.png

Insight

Japanese and European cars are noticeably more fuel-efficient than US-made cars, which tend to be heavier.

Chart 4

MPG by Cylinders (Bar Chart) → chart4_mpg_by_cylinders.png

Insight

Average mpg drops sharply as cylinder count increases — 4-cylinder cars are far more efficient than 8-cylinder cars.

Chart 5

Correlation Heatmap (Heatmap) → chart5_correlation_heatmap.png

Insight

Weight (-0.83) and displacement (-0.80) show the strongest negative correlation with mpg, confirming them as the top predictors.

6 Model Training & Comparison

Setup

Setting	Value
Features Used	cylinders, displacement, horsepower, weight, acceleration, model_year, origin_num
Target	mpg
Train / Test Split	80% / 20% (random_state=42)
Models Trained	Linear Regression & Random Forest Regressor (100 trees)

```
model1 = LinearRegression()
model1.fit(X_train, y_train)

model2 = RandomForestRegressor(n_estimators=100, random_state=42)
model2.fit(X_train, y_train)
```

Results

Model	MAE	RMSE	R2
Linear Regression	2.26	2.86	0.85
Random Forest Regressor	1.58	2.14	0.91

Insight Random Forest outperformed Linear Regression on every metric — lower MAE (1.58 vs 2.26), lower RMSE (2.14 vs 2.86), and higher R2 (0.91 vs 0.85). It explains 91% of the variance in mpg, compared to 85% for the linear model. This suggests mpg depends on features in a non-linear way that tree-based models capture better.

7 Key Findings & Conclusion

- Car weight is the single strongest predictor of fuel efficiency (correlation: -0.83)
- Random Forest beat Linear Regression on every metric — MAE, RMSE, and R2
- Random Forest explains 91% of mpg variance vs 85% for Linear Regression
- Tree-based models capture non-linear relationships that linear models miss
- Japanese and European cars are consistently more fuel-efficient than US cars
- Fewer engine cylinders strongly correlates with better fuel efficiency

Task Status: Complete — All 5 objectives met. Dataset preprocessed and feature-engineered, 5-chart dashboard created, two regression models trained and compared with MAE/RMSE/R2, and findings documented. Random Forest selected as the best-performing model. Project structured and pushed to GitHub.